PUPIL: _____ CLASS: _____ TEACHER: _____

KEY TOPIC 5 • 6 ENQUIRY HOURS • 1914–1918

The British Sector of the Western Front, 1914–1918

Injuries, Treatment and the Trenches in the British Sector



Primary Visual Evidence: *Stretcher Bearers of the RAMC Lifting a Wounded Man out of a Trench* • Gilbert Rogers (c. 1919)

OFFICIAL EDEXCEL SOURCE ARCHIVE • IWM ART 2485 / RAMC HISTORIC COLLECTION

1. FEATURE DRILLS [4M]
Q1(a) & Q1(b): Two 2-mark questions. Identify 1 specific feature + 1 supporting factual detail (5 mins).

2. SOURCE UTILITY [8M]
Q2(a): 3-step evaluation of Content, Provenance (Nature, Origin, Motive), and Contextual Limitations (10 mins).

3. FOLLOW-UP ENQUIRY [4M]
Q2(b): 4-row tabular enquiry: Detail, Historical Question, Source Type, and Clinical Value (5 mins).

COURSE SPECIFICATION • KEY ENQUIRY SEQUENCE (1914–1918)	LEARNT	REVISED
Key Topic 5.1: The Theatre of War: The British Sector, Trench Geography & Battles *Why were trenches designed in zig-zag traverses, and how did terrain at Ypres, Somme, and Arras affect medical treatment?*	☐	☐
Key Topic 5.2: The Trench Environment: Mud, Vermin & Non-Combat Illnesses *How did frontline conditions cause trench foot, trench fever, and dysentery, and what preventive routines were enforced?*	☐	☐
Key Topic 5.3: Battlefield Trauma: High Explosive Shrapnel, Gas Attacks & Infection *Why did explosive artillery produce gas gangrene, and how did the Brodie steel helmet and respirators reduce fatalities?*	☐	☐
Key Topic 5.4: The Chain of Evacuation: Stretcher Bearers, RAP, Dressing Stations & CCS *How did the RAMC and FANY evacuate wounded soldiers through six stations, and why was triage critical at CCSs?*	☐	☐
Key Topic 5.5: Surgical Breakthroughs: The Thomas Splint, Wound Debridement & Mobile X-Rays *How did the Thomas Splint slash femur mortality from 80% to 20%, and why was Carrel-Dakin irrigation necessary?*	☐	☐
Key Topic 5.6: Lifesaving Innovations: Blood Storage, Brain Surgery & Plastic Reconstruction *How did Oswald Robertson establish the first blood bank at Cambrai, and how did Gillies rebuild shattered faces at Sidcup?*	☐	☐

LIVING TIMELINE • PART 1: FRONTLINE CONDITIONS & EARLY TRAUMA (1914–1915)

Instructions: As you study each enquiry lesson, complete the timeline missions by sketching and annotating in the corresponding Key Topic boxes below.

OCT 1914 • Flanders Mud & The 1st Battle of Ypres

Key Topic 5.1

The British Expeditionary Force (BEF) holds the Ypres Salient. Low-lying Flanders clay creates a dangerously high water table; trenches fill with liquid mud. Shelling smashes all natural drainage canals, leaving men standing waist-deep in water and giving rise to the first epidemics of trench foot and dysentery.

APR 1915 • 2nd Battle of Ypres: The First Poison Gas Attack

Key Topic 5.2 • 5.3

German forces release 160 tons of chlorine gas across a 4-mile front. British troops suffer suffocation and fluid in the lungs; soldiers improvise with urine-soaked pads before chemically impregnated cotton-pad respirators (the Hypo Helmet) are rushed to the front. Lice-borne trench fever breaks out across all divisions.

DEC 1915 • The Thomas Splint & Introduction of the Brodie Helmet

Key Topic 5.3 • 5.5

Robert Jones introduces the Thomas Splint to frontline medical posts, stabilizing compound fractures and reducing femur mortality from 80% to 20%. Simultaneously, the War Office begins issuing John Brodie's stamped steel helmet, reducing penetrating shrapnel head trauma by an estimated 75–80%.

If you think your walk to period 1 is rough, try carrying a 14-stone sergeant through 4 miles of knee-deep Flanders clay.

LIVING TIMELINE • PART 2: MASS OFFENSIVES & CLINICAL INNOVATION (1916–1917)

Instructions: Complete the timeline missions by sketching and annotating in the corresponding Key Topic boxes below.

JULY 1916 • The Battle of the Somme: The Triage Crisis

Key Topic 5.4

The BEF suffers 57,000 casualties on 1 July 1916. Stretcher bearers and Casualty Clearing Stations are overwhelmed; motor ambulances struggle along single rutted roads. As a direct result, the RAMC enforces a strict three-tier triage system (Walking Wounded, Immediate Surgery, Moribund) to prevent total collapse.

APR–NOV 1917 • Arras Underground Hospital & Passchendaele Mud

Key Topic 5.1 • 5.4

At Arras, New Zealand tunnelers excavate Thompson's Cave—a 700-bed underground hospital with operating theatres, running water, and electric lights 20m beneath the chalk. At 3rd Ypres (Passchendaele), artillery destroys all drainage systems, creating knee-deep liquid mud requiring 6 bearers per stretcher.

NOV 1917 • Oswald Robertson's Blood Depot & Gillies Facial Reconstruction

Key Topic 5.6

At the Battle of Cambrai, Captain Oswald Robertson deploys the world's first blood bank, using sodium citrate to stop clotting, glucose to preserve red cells, and ice chests to store Group O blood for 28 days. In Britain, Harold Gillies pioneers plastic facial surgery using tubed pedicles at Queen's Hospital, Sidcup.

How did the physical terrain and trench architecture of the British sector dictate medical evacuation and care?

Specification Requirement: *The British sector of the Western Front: theatre of war in Flanders and northern France; Ypres, Somme, Arras, Cambrai; trench system (frontline, support, reserve, communication trenches; saps); terrain, clay, water table, mud; underground systems at Arras (Thompson's Cave).*

1 **1914–1915**

Flanders Terrain & Ypres Salient

- Low-lying Flanders clay: very high water table
- Heavy rain and artillery blast destroy drainage
- British held vulnerable salient overlooked by Germans
- Liquid mud made stretcher evacuation agonizingly slow

FOCUS CLUE
Why was the low-lying topography of Ypres a tactical and medical nightmare?

2 **1914–1918**

Trench Architecture & Traverses

- Trenches dug 2m deep in zig-zag traverse pattern
- Traverses contain shell blast and flying shrapnel
- Prevents enemy firing straight down length of trench
- Duckboards, fire-steps, parapets, and parados fitted

FOCUS CLUE
How did zig-zag traverses limit shrapnel injuries and contain shell explosions?

3 **1915–1918**

Trench System: Front to Rear

- Frontline: fire-steps for observation and combat
- Support trench (70m back) & Reserve (hundreds m)
- Communication trenches move fresh men & stretchers
- Saps: narrow blind tunnels into No Man's Land

FOCUS CLUE
Why did narrow communication trenches create acute bottlenecks for stretchers?

4 **Nov 1916–1917**

Thompson's Cave & Arras Tunnels

- Chalk geology enabled massive tunnel excavations
- New Zealand tunnelers joined ancient Roman quarries
- Electric lighting, running piped water & rail tracks
- 700-bed underground hospital (Thompson's Cave)

FOCUS CLUE
How did the chalk geology at Arras allow safe, sterile surgical conditions?

5 **1916–1917**

Mass Trauma: Somme & Cambrai

- July 1916 Somme: 57,000 British casualties Day 1
- Stretcher bearers overwhelmed by mass casualties
- Nov 1917 Cambrai: mass tank assault over chalk ridge
- Mobile breakthrough demanded rapid surgical shifts

FOCUS CLUE
Why did mass-casualty offensives force the RAMC to reorganize evacuation?

TIMELINE MISSION **Turn to Pages 2–3 (Milestone 1). Sketch and annotate the zig-zag traverse pattern and low-lying Ypres water table.** ← Pages 2–3

Welcome to Flanders: 500 square miles of fermented pig manure, liquid mud, and artillery with terrifying accuracy.

KT5.1: The Theatre of War: The British Sector, Trench Geography & Battles

SOURCE A • CONTEMPORARY AERIAL RECONNAISSANCE • ROYAL FLYING CORPS **ORIGIN: 1917**



Royal Flying Corps aerial photograph showing the intricate zig-zag traverses of frontline, support, and communication trenches near Ypres, criss-crossed with artillery shell craters.

Historical Provenance: Official military aerial photograph taken by the Royal Flying Corps, British Sector near Ypres, 1917. Imperial War Museum Collection.

• EDEXCEL SECTION A: QUESTION 1 FEATURE DRILLS [4 MARKS • 5 MINS] **2 SEPARATE QUESTIONS • 2 MARKS EACH**

Examiner Mark Scheme: Award 1 mark for identifying a valid historical feature + 1 mark for supporting factual detail. To score 2/2, the supporting detail must directly explain and develop the named feature.

1(a) Describe one feature of the system of communication trenches on the Western Front. [2 marks]
Target: AO1 [2 marks] • Point (1m) + Detail (1m)
Sentence Stems: *One key feature was that... • Specifically, this was evident when...*

1(b) Describe one feature of the underground hospital at Arras (Thompson's Cave). [2 marks]
Target: AO1 [2 marks] • Point (1m) + Detail (1m)
Sentence Stems: *One key feature was that... • Specifically, this was evident when...*

• QUESTION 2(A): SOURCE UTILITY ASSESSMENT [8 MARKS • 10 MINS] **AO3 EVALUATION**

Study Source A. How useful is Source A for an enquiry into the defensive layout and tactical design of trench systems on the Western Front? Explain your answer, using Source A and your knowledge of the historical context. [8 marks]

Provenance Clue: Consider the nature of an official military aerial reconnaissance photograph taken by the Royal Flying Corps in 1917. What does an overhead view reveal about traverses, fire-steps, and saps, and what ground-level medical and mud conditions does it omit?

ANALYTICAL CONNECTIVES:
Source A is useful for an enquiry into this topic because... • Specifically, the source reveals... • In terms of provenance, considering the author and date... • However, the utility is limited because... • Therefore...

KEY VOCABULARY:
• Royal Flying Corps • zig-zag traverses • communication trench • Ypres Salient • Messines Ridge • water table • duckboards • parapet • parados • fire-step

SOURCE EVIDENCE, CONTEXTUAL KNOWLEDGE & DISCIPLINARY ANALYSIS NOTES:

FORMAL EXAMINATION PRACTICE **Write your formal answers on Pages 6–7 →**

• QUESTION 1: FEATURE DRILLS [2 X 2 = 4 MARKS] **Q1 Total: [/ 4]**

1(a) Describe one feature of the system of communication trenches on the Western Front. **[/ 2]**

FEATURE: _____

DETAIL: _____

1(b) Describe one feature of the underground hospital at Arras (Thompson's Cave). **[/ 2]**

FEATURE: _____

DETAIL: _____

• QUESTION 2(A): SOURCE UTILITY RESPONSE [8 MARKS • 10 MINS] Continue on next page →

[Begin Question 2(a) Utility Evaluation • Content • Provenance • Context]

Ypres Salient: surrounded on three sides by German artillery on Messines Ridge; duckboards were life preservers.

[Question 2 Sustained Analysis Continued • Contextual Knowledge • Conclusion]

Q2 Score: [/ 8] Lesson Total: [/ 12] Teacher Feedback / Target: .....

Why were environmental pathogens and vermin as dangerous to British troops as enemy artillery?

Specification Requirement: *Ill health arising from the trench environment: trench foot (pathology, gangrene, prevention, whale oil, sock inspection); trench fever (body lice, Pediculus humanus, delousing stations, Serbian barrels); shell shock (symptoms, NYDN, rest treatment); dysentery (water chlorination, chloride of lime, latrines); underground shelter conditions.*

1 **1914–1915**

Trench Foot Pathology

- Caused by standing in cold water and mud in tight boots
- Constricts capillary blood flow; tissues lose oxygen
- Numbness, severe swelling, skin turning blue and black
- Led to wet gangrene, tissue necrosis and amputation

FOCUS CLUE
How did prolonged cold water immersion destroy tissue and produce gangrene?

2 **1915–1918**

Prevention: Whale Oil & Socks

- Mandatory buddy system: soldiers rubbed whale oil daily
- Whale oil formed water-repellent barrier on skin
- Daily foot inspections strictly enforced by officers
- Every soldier carried three pairs of clean, dry socks

FOCUS CLUE
Why did mandatory officer inspections succeed where voluntary advice had failed?

3 **1914–1918**

Trench Fever, Lice & Delousing

- High fever, violent shivering, severe joint and shin pain
- Incapacitated up to 15% of troops for a month or more
- 1918: Proven transmitted by body lice (Pediculus humanus) faeces
- Divisional delousing bathhouses, steam ovens & Serbian barrels

FOCUS CLUE
Why did trench fever persist until the body louse insect vector was proven in 1918?

4 **1914–1918**

Dysentery & Water Chlorination

- Severe bacterial diarrhoea caused by unhygienic water & food
- Deep latrine pits dug away from frontline trenches
- Chloride of lime added daily to neutralize foul waste
- Drinking water purified with chemical chlorination tablets

FOCUS CLUE
How did strict sanitation and chemical water chlorination prevent epidemics?

5 **1914–1918**

Shell Shock & Psychological Trauma

- Caused by artillery blast concussion and mortal combat terror
- Symptoms: uncontrollable shaking, hysterical blindness, mutism
- ~80,000 British cases diagnosed; labeled NYDN (Not Yet Diagnosed, Nervous)
- Treated with rest and food; specialist centres like Craiglockhart

FOCUS CLUE
Why was shell shock initially stigmatised as cowardice before being treated medically?

ges 2–3 (Milestone 2). Draw soldiers applying whale oil and delousing garments in steam ovens. ← Pages 2–3

Whale oil smells like dead fish and regrets, but it beats having your toes amputated by an RAMC surgeon with a bone-saw.

KT5.2: The Trench Environment: Mud, Vermin & Non-Combat Illnesses

SOURCE A • CONTEMPORARY PHOTOGRAPHIC EVIDENCE • WESTERN FRONT **ORIGIN: July 1916**



British infantrymen of 'A' Company, 11th Battalion, Cheshire Regiment, resting wrapped in waterproof groundsheets on the duckboards and dugouts of a captured trench at Ovillers-la-Boisselle, while a sentry remains on duty on the parapet.

Historical Provenance: Official British War Office photograph taken by Lieutenant John Warwick Brooke during the Battle of the Somme, July 1916. Imperial War Museum Collection.

• EDEXCEL SECTION A: QUESTION 1 FEATURE DRILLS [4 MARKS • 5 MINS] **2 SEPARATE QUESTIONS • 2 MARKS EACH**

Examiner Mark Scheme: Award 1 mark for identifying a valid historical feature + 1 mark for supporting factual detail. To score 2/2, the supporting detail must directly explain and develop the named feature.

1(a) Describe one feature of the methods used to prevent trench foot on the Western Front. [2 marks]
Target: AO1 [2 marks] • Point (1m) + Detail (1m)
Sentence Stems: *One key feature was that... • Specifically, this was evident when...*

1(b) Describe one feature of the causes of trench fever among British troops. [2 marks]
Target: AO1 [2 marks] • Point (1m) + Detail (1m)
Sentence Stems: *One key feature was that... • Specifically, this was evident when...*

• QUESTION 2(B): 4-PROMPT FOLLOW-UP ENQUIRY [4 MARKS • 5 MINS] **AO3 HISTORICAL ENQUIRY**

Study Source A. How could you follow up Source A to find out more about the living conditions and non-combat illnesses experienced by British soldiers in frontline trenches on the Western Front? [4 marks]

Examiner Strategy: Follow up an exact quotation or visual detail. Your question must directly target your chosen detail. Your named source type must be authentic and specific (e.g. RAMC casualty returns, battalion war diaries).

Official Edexcel Prompt	AO3 Disciplinary Enquiry Strategy • Exam Standard
1. Detail in Source A to follow up:	*Select a precise, verifiable fact or visual detail directly related to the enquiry.*
2. Question I would ask:	*Frame a focused historical question that directly investigates your chosen detail.*
3. Type of source I would use:	*Name a specific contemporary record (e.g. RAMC war diaries, casualty logs, medical records).*
4. How this would help me:	*Explain what unique statistical or medical data this source provides to answer your question.*

Key Vocabulary: trench foot • whale oil • buddy-system • gangrene • trench fever • body lice (Pediculus humanus) • delousing stations • Serbian barrels • shell shock • NYDN • dysentery • chloride of lime

A key detail to follow up is... • The question I would ask is... • The specific historical source type needed is... • This source would help answer my question because...

SOURCE EVIDENCE, CONTEXTUAL KNOWLEDGE & DISCIPLINARY ANALYSIS NOTES:

FORMAL EXAMINATION PRACTICE **Write your formal answers on Pages 10–11 →**

• QUESTION 1: FEATURE DRILLS [2 X 2 = 4 MARKS] **Q1 Total: [/ 4]**

1(a) Describe one feature of the methods used to prevent trench foot on the Western Front. **[/ 2]**

FEATURE: _____

DETAIL: _____

_____

1(b) Describe one feature of the causes of trench fever among British troops. **[/ 2]**

FEATURE: _____

DETAIL: _____

_____

• QUESTION 2(B): 4-PROMPT FOLLOW-UP MATRIX [4 MARKS • 5 MINS] **Score: [/ 4]**

Detail in Source A that I would follow up:	
Question I would ask:	
Type of source I would use:	
How this would help me find out more:	

[Disciplinary Commentary: Justify why this source type is authoritative for your historical enquiry]

Trench foot required 3 pairs of dry socks and rubbing with whale oil; negligence resulted in court martial.

[Question 2 Sustained Analysis Continued • Contextual Knowledge • Conclusion]

Q2 Score: [/ 4] **Lesson Total: [/ 8]** **Teacher Feedback / Target:** .....

How did high-explosive artillery trauma and chemical poison gas transform surgical treatment?

Specification Requirement: *Wounds, injuries and diseases: high-explosive artillery shells, shrapnel fragments; infection (gas gangrene, Clostridium welchii, tetanus); head trauma and the Brodie helmet; chemical gas attacks (chlorine, phosgene, mustard gas) and respirators.*

1 **1914–1918**

Artillery & Shrapnel Trauma

- High-explosive artillery caused ~58% of all battle wounds
- Shells shattered into thousands of jagged steel splinters
- Smashed complex bones, caused blast shock and severe bleeding
- Mud, dirt, and filthy uniform cloth driven deep into tissue

FOCUS CLUE
Why did jagged shrapnel fragments cause far worse infections than rifle bullets?

2 **1914–1918**

Soil Bacteria & Gas Gangrene

- Manured Flanders farmland heavily contaminated with microbes
- Clostridium welchii bacterium produces lethal toxins and gas
- Anaerobic: flourishes in deep, airless shrapnel wounds
- Gas bubbles under skin; gangrene killed soldiers in hours

FOCUS CLUE
Why was Clostridium welchii so lethal in deep, airless shrapnel wounds?

3 **Late 1915**

Head Trauma & Brodie Helmet

- Soft cloth caps offered zero protection against shrapnel
- Late 1915: John Brodie patented the steel combat helmet
- Stamped from single steel sheet; wide rim deflected blast
- Reduced fatal penetrating shrapnel head wounds by ~80%

FOCUS CLUE
Why did the introduction of the Brodie helmet cause an apparent rise in hospital head cases?

4 **Apr 1915–1916**

Chlorine & Phosgene Attacks

- Apr 1915: Germans used chlorine gas at 2nd Battle of Ypres
- Choking gas destroyed respiratory lungs; fluid suffocation
- Urine pads replaced by chemical hypo hoods and P-helmets
- Dec 1915: Phosgene introduced (faint musty smell; 6x deadlier)

FOCUS CLUE
How did the rapid evolution of chemical respirators counter suffocating gases?

5 **1916–1918**

Mustard Gas & Box Respirators

- 1916: Small Box Respirator issued (carbon/charcoal filter)
- July 1917: Germans deployed mustard gas at 3rd Ypres
- Blistering agent burned skin through uniforms; caused blindness
- Odorless, oily liquid soaked into soil; persisted for weeks

FOCUS CLUE
Why was mustard gas far harder to counter than chlorine or phosgene?

TIMELINE MISSION **Turn to Pages 2–3 (Milestone 2). Sketch the Brodie steel helmet and British PH gas hood.** ← Pages 2–3

The Brodie helmet: Looks like an upside-down soup bowl, but prevents your skull from becoming one.

KT5.3: Battlefield Trauma: High Explosive Shrapnel, Gas Attacks & Infection

SOURCE B • CONTEMPORARY PHOTOGRAPHIC EVIDENCE • MACHINE GUN CORPS **ORIGIN: July 1916**



A British Machine Gun Corps crew operating a Vickers machine gun from an improvised trench position near Ovillers on the Somme while wearing issue Phenate-Hexamine (PH) anti-gas helmets with exhale valves.

Historical Provenance: Official British War Office photograph taken by Lieutenant John Warwick Brooke near Ovillers during the Battle of the Somme, July 1916. Imperial War Museum Collection.

• EDEXCEL SECTION A: QUESTION 1 FEATURE DRILLS [4 MARKS • 5 MINS] **2 SEPARATE QUESTIONS • 2 MARKS EACH**

Examiner Mark Scheme: Award 1 mark for identifying a valid historical feature + 1 mark for supporting factual detail. To score 2/2, the supporting detail must directly explain and develop the named feature.

1(a) Describe one feature of the effects of poison gas attacks on soldiers on the Western Front. [2 marks]
Target: AO1 [2 marks] • Point (1m) + Detail (1m)
Sentence Stems: *One key feature was that... • Specifically, this was evident when...*

1(b) Describe one feature of the design of the British Brodie steel helmet. [2 marks]
Target: AO1 [2 marks] • Point (1m) + Detail (1m)
Sentence Stems: *One key feature was that... • Specifically, this was evident when...*

• QUESTION 2(A): SOURCE UTILITY ASSESSMENT [8 MARKS • 10 MINS] **AO3 EVALUATION**

Study Source B. How useful is Source B for an enquiry into the methods used to protect British soldiers from poison gas attacks on the Western Front? Explain your answer, using Source B and your knowledge of the historical context. [8 marks]

Provenance Clue: Consider the nature of an authentic British War Office Phenate-Hexamine (PH) anti-gas helmet manufactured in 1915–1916. What physical protective properties does chemically impregnated flannel possess, and what severe combat limitations did it have in battle?

ANALYTICAL CONNECTIVES:
Source B is useful for investigating this topic because... • Specifically, the source provides evidence that... • In terms of provenance, considering the nature and purpose... • However, the source is limited because... • Consequently...

KEY VOCABULARY:
high-explosive artillery • shrapnel balls • gas gangrene • Clostridium welchii • Brodie steel helmet • chlorine gas (1915) • phosgene • mustard gas (1917) • PH helmet • Small Box Respirator (1916)

SOURCE EVIDENCE, CONTEXTUAL KNOWLEDGE & DISCIPLINARY ANALYSIS NOTES:

FORMAL EXAMINATION PRACTICE **Write your formal answers on Pages 14–15 →**

• QUESTION 1: FEATURE DRILLS [2 X 2 = 4 MARKS] **Q1 Total: [/ 4]**

1(a) Describe one feature of the effects of poison gas attacks on soldiers on the Western Front. **[/ 2]**

FEATURE: _____

DETAIL: _____

1(b) Describe one feature of the design of the British Brodie steel helmet. **[/ 2]**

FEATURE: _____

DETAIL: _____

• QUESTION 2(A): SOURCE UTILITY RESPONSE [8 MARKS • 10 MINS] Continue on next page →

[Begin Question 2(a) Utility Evaluation • Content • Provenance • Context]

The 1915 Brodie helmet slashed penetrating head wounds by 75%; shrapnel helmets became universal equipment.

[Question 2 Sustained Analysis Continued • Contextual Knowledge • Conclusion]

Q2 Score: [/ 8] Lesson Total: [/ 12] Teacher Feedback / Target: .....

How successfully did the Chain of Evacuation transport, triage, and treat mass battlefield casualties?

Specification Requirement: *The work of the RAMC and FANY: the Chain of Evacuation (stretcher bearers, Regimental Aid Posts, Field Ambulances and Dressing Stations, Casualty Clearing Stations, Base Hospitals); triage systems; transport methods (motor ambulances, ambulance trains, canal barges).*

1 **1914–1918**

Stretcher Bearers & RAP

- 16 bearers per battalion (4 per stretcher; 6–8 in mud)
- Recovered wounded under direct machine-gun/artillery fire
- Regimental Aid Post (RAP) 200m behind front line in dugout
- Regimental Medical Officer (RMO) applied iodine & bandages

FOCUS CLUE
Why was rapid evacuation by stretcher bearers vital to prevent irreversible shock?

2 **1914–1918**

Field Ambulances & ADS/MDS

- Staffed by Royal Army Medical Corps (RAMC) Field Ambulance
- Advanced Dressing Station (1/2 mile back); MDS (1 mile back)
- Located in abandoned farmhouses, church ruins, or bunkers
- Minor wounds dressed; severe cases dispatched by ambulance

FOCUS CLUE
What was the tactical role of Dressing Stations in preventing CCS congestion?

3 **1914–1918**

CCS & The Triage System

- Located 7–12 miles behind front, near railway junctions
- Primary surgical centre for critical life-saving operations
- Triage: 1. Walking Wounded, 2. Immediate, 3. Moribund
- Major amputations and emergency abdominal laparotomies done

FOCUS CLUE
How did the 3-tier triage system maximize survival during mass offensives?

4 **1914–1918**

Base Hospitals on French Coast

- Located at coastal ports (Boulogne, Calais, Le Havre, Étaples)
- Large civilian hospitals, hotels, or vast tent cities
- Specialized wards: amputees, gas burns, chest injuries
- Patients convalesced or shipped back to Britain on hospital ships

FOCUS CLUE
Why were Base Hospitals situated near major coastal rail terminals and ports?

5 **1914–1918**

Motor Ambulances & Barges

- Horse ambulances caused agony; shaken by rough shell-holes
- 1914 Red Cross appeal raised 512 motorized ambulances
- Hospital trains fitted with kitchens and operating rooms
- Canal barges provided smooth travel for head and chest trauma

FOCUS CLUE
Why were canal barges preferred over trains for head and chest trauma?

TIMELINE MISSION **Turn to Pages 2–3 (Milestone 5). Draw the 6 sequential stages of the Chain of Evacuation.** ← Pages 2–3

The Regimental Aid Post: 200 yards from the German front line, lit by candle-ends, and smelling entirely of iodine and panic.

KT5.4: The Chain of Evacuation: Stretcher Bearers, RAP, Dressing Stations & CCS

SOURCE A • CONTEMPORARY PHOTOGRAPHIC EVIDENCE • WESTERN FRONT **ORIGIN: August 1917**



A team of six British stretcher bearers struggling through deep, liquid Flanders mud to carry a wounded soldier on a wooden stretcher near Boesinghe during the Third Battle of Ypres.

Historical Provenance: Official British Army photograph taken by Lieutenant John Warwick Brooke near Boesinghe during the Third Battle of Ypres, 1 August 1917. Imperial War Museum Collection.

• EDEXCEL SECTION A: QUESTION 1 FEATURE DRILLS [4 MARKS • 5 MINS] **2 SEPARATE QUESTIONS • 2 MARKS EACH**

Examiner Mark Scheme: Award 1 mark for identifying a valid historical feature + 1 mark for supporting factual detail. To score 2/2, the supporting detail must directly explain and develop the named feature.

1(a) Describe one feature of the triage system used at Casualty Clearing Stations on the Western Front. [2 marks]
Target: AO1 [2 marks] • Point (1m) + Detail (1m)
Sentence Stems: *One key feature was that... • Specifically, this was evident when...*

1(b) Describe one feature of the work of the First Aid Nursing Yeomanry (FANY) on the Western Front. [2 marks]
Target: AO1 [2 marks] • Point (1m) + Detail (1m)
Sentence Stems: *One key feature was that... • Specifically, this was evident when...*

• QUESTION 2(B): 4-PROMPT FOLLOW-UP ENQUIRY [4 MARKS • 5 MINS] **AO3 HISTORICAL ENQUIRY**

Study Source A. How could you follow up Source A to find out more about the difficulties stretcher bearers faced evacuating casualties from the Western Front? [4 marks]

Examiner Strategy: Follow up an exact quotation or visual detail. Your question must directly target your chosen detail. Your named source type must be authentic and specific (e.g. RAMC casualty returns, battalion war diaries).

Official Edexcel Prompt	AO3 Disciplinary Enquiry Strategy • Exam Standard
1. Detail in Source A to follow up:	*Select a precise, verifiable fact or visual detail directly related to the enquiry.*
2. Question I would ask:	*Frame a focused historical question that directly investigates your chosen detail.*
3. Type of source I would use:	*Name a specific contemporary record (e.g. RAMC war diaries, casualty logs, medical records).*
4. How this would help me:	*Explain what unique statistical or medical data this source provides to answer your question.*

Key Vocabulary: RAMC • FANY • Regimental Aid Post • Advanced Dressing Station • Casualty Clearing Station • Base Hospital • triage system • motor ambulance • hospital train • canal barge

A key detail to follow up is... • The question I would ask is... • The specific historical source type needed is... • This source would help answer my question because...

SOURCE EVIDENCE, CONTEXTUAL KNOWLEDGE & DISCIPLINARY ANALYSIS NOTES:

FORMAL EXAMINATION PRACTICE **Write your formal answers on Pages 18–19 →**

• QUESTION 1: FEATURE DRILLS [2 X 2 = 4 MARKS] **Q1 Total: [/4]**

1(a) Describe one feature of the triage system used at Casualty Clearing Stations on the Western Front. **[/2]**

FEATURE: _____

DETAIL: _____

_____

1(b) Describe one feature of the work of the First Aid Nursing Yeomanry (FANY) on the Western Front. **[/2]**

FEATURE: _____

DETAIL: _____

_____

• QUESTION 2(B): 4-PROMPT FOLLOW-UP MATRIX [4 MARKS • 5 MINS] **Score: [/4]**

Detail in Source A that I would follow up:	
Question I would ask:	
Type of source I would use:	
How this would help me find out more:	

[Disciplinary Commentary: Justify why this source type is authoritative for your historical enquiry]

[Question 2 Sustained Analysis Continued • Contextual Knowledge • Conclusion]

Q2 Score: [/ 4] Lesson Total: [/ 8] Teacher Feedback / Target: .....

Why were the Thomas Splint, debridement, and mobile X-rays decisive surgical breakthroughs?

Specification Requirement: *Medical advances on the Western Front: the Thomas Splint (Hugh Owen Thomas, Robert Jones) reducing compound femur mortality; wound debridement and delayed primary closure; the Carrel-Dakin antiseptic irrigation method; mobile X-ray units and radiology.*

1 **1914–1918**

Wound Debridement & Excision

- Surface antiseptics failed against deep dirty shrapnel wounds
- Surgeons cut away all dead, bruised, and infected tissue
- Deprived anaerobic bacteria (Clostridium welchii) of dead flesh
- Delayed primary closure: wound left open to flush before stitching

FOCUS CLUE
Why was surgical excision of dead tissue more effective than chemical [illegible]?

2 **1915–1918**

Carrel-Dakin Antiseptic Flush

- Alexis Carrel & Henry Dakin design continuous wound irrigation
- Dilute sodium hypochlorite (buffered chlorine solution)
- Flushed into deep wound through network of perforated rubber tubes
- Prevented gangrene; degraded quickly so made fresh every 24h

FOCUS CLUE
How did continuous chemical irrigation solve the problem of deep wound infection?

3 **Dec 1915–1916**

The Thomas Splint Breakthrough

- Compound femur fracture mortality in 1914 stood at 80%
- Muscle spasms pulled sharp broken bone ends through femoral artery
- Hugh Owen Thomas splint introduced to front by Robert Jones
- Secured leg in rigid mechanical traction; mortality dropped to 20%

FOCUS CLUE
How did rigid traction prevent fatal haemorrhage and shock from broken femurs?

4 **1914–1918**

Mobile X-Rays & Marie Curie

- Essential to locate shrapnel and bullets before surgical incision
- Marie Curie equipped and deployed 20 mobile X-ray vans
- Powered by van engines; operated near Casualty Clearing Stations
- Static base hospital X-rays located deep fragments and gas pockets

FOCUS CLUE
Why was locating shrapnel on an X-ray vital before operating on gas gangrene?

5 **1914–1918**

X-Ray Limits in Combat

- Glass X-ray tubes fragile; easily shattered on rutted roads
- Tubes overheated after minutes; required cooling periods
- Could not detect cloth, mud, or wooden splinters in tissue
- Severely wounded soldiers had to lie completely still for minutes

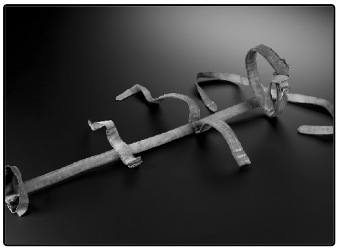
FOCUS CLUE
Why did undetected cloth fragments remain the primary cause of post-op gangrene?

TIMELINE MISSION **Turn to Pages 2–3 (Milestone 3). Sketch the Thomas Splint groin ring and traction cord.** ← Pages 2–3

The Thomas Splint: Before Robert Jones introduced it, an 80% chance of death; after Jones, an 80% chance of living to complain about the food.

KT5.5: Surgical Breakthroughs: The Thomas Splint, Wound Debridement & Mobile X-Rays

SOURCE A • CONTEMPORARY MEDICAL ARTIFACT • STANDARD ISSUE **ORIGIN: c.1916**



An authentic surviving British Army Thomas Splint, showing the padded oval ring fitted against the groin, rigid metal side bars, and canvas support straps used to maintain traction and immobilize broken thigh bones.

Historical Provenance: Original First World War Thomas Splint preserved in the Wellcome Historical Medical Collection, London. Designed by Hugh Owen Thomas and introduced to the Western Front in 1915–1916 by Robert Jones.

• EDEXCEL SECTION A: QUESTION 1 FEATURE DRILLS [4 MARKS • 5 MINS] **2 SEPARATE QUESTIONS • 2 MARKS EACH**

Examiner Mark Scheme: Award 1 mark for identifying a valid historical feature + 1 mark for supporting factual detail. To score 2/2, the supporting detail must directly explain and develop the named feature.

1(a) Describe one feature of the Thomas Splint used on the Western Front. [2 marks]

Target: AO1 [2 marks] • Point (1m) + Detail (1m)
Sentence Stems: *One key feature was that... • Specifically, this was evident when...*

1(b) Describe one feature of the Carrel-Dakin method of treating infected wounds. [2 marks]

Target: AO1 [2 marks] • Point (1m) + Detail (1m)
Sentence Stems: *One key feature was that... • Specifically, this was evident when...*

• QUESTION 2(A): SOURCE UTILITY ASSESSMENT [8 MARKS • 10 MINS] **AO3 EVALUATION**

Study Source A. How useful is Source A for an enquiry into the methods used to treat wounded soldiers with fractured limbs on the Western Front? Explain your answer, using Source A and your knowledge of the historical context. [8 marks]

Provenance Clue: Consider the nature of an authentic instructional photograph taken in an RAMC military hospital training ward c.1916. What does it demonstrate about mechanical traction and splinting, and why might an instructional training photograph differ from emergency frontline conditions in a mud dugout?

ANALYTICAL CONNECTIVES:

Source A is useful for an enquiry into this topic because... • Specifically, the source demonstrates that... • In terms of provenance, considering the contemporary context... • However, the utility is limited because... • Therefore...

KEY VOCABULARY:

Thomas Splint • Hugh Owen Thomas • Robert Jones • compound fracture • femoral artery • mechanical traction • wound debridement • Carrel-Dakin solution • delayed primary closure • mobile X-ray unit

SOURCE EVIDENCE, CONTEXTUAL KNOWLEDGE & DISCIPLINARY ANALYSIS NOTES:

FORMAL EXAMINATION PRACTICE **Write your formal answers on Pages 22–23 →**

• QUESTION 1: FEATURE DRILLS [2 X 2 = 4 MARKS] **Q1 Total: [/ 4]**

1(a) Describe one feature of the Thomas Splint used on the Western Front. **[/ 2]**

FEATURE: _____

DETAIL: _____

1(b) Describe one feature of the Carrel-Dakin method of treating infected wounds. **[/ 2]**

FEATURE: _____

DETAIL: _____

• QUESTION 2(A): SOURCE UTILITY RESPONSE [8 MARKS • 10 MINS] **Continue on next page →**

[Begin Question 2(a) Utility Evaluation • Content • Provenance • Context]

[Question 2 Sustained Analysis Continued • Contextual Knowledge • Conclusion]

Q2 Score: [/ 8] Lesson Total: [/ 12] Teacher Feedback / Target: .....

How did stored blood banks, specialized neurosurgery, and plastic reconstruction transform wartime survival?

Specification Requirement: *Medical advances on the Western Front: blood transfusions and storage (Landsteiner, Hustin, Rous and Turner, Captain Oswald Robertson and the Cambrai blood bank); specialized neurosurgery (Harvey Cushing); plastic and facial reconstruction (Harold Gillies, Queen's Hospital Sidcup, tubed pedicle).*

1 **1914–1915**

Sodium Citrate & Clotting

- 1914: Albert Hustin discovers sodium citrate stops clotting
- Prevents blood from coagulating when exposed to air
- 1915: Richard Lewisohn determines safe non-toxic dosage (0.2%)
- Enabled indirect transfusion: donor no longer strapped to patient

FOCUS CLUE
How did sodium citrate liberate blood transfusion from donor-to-patient tubing?

2 **1916–1917**

Citrate-Glucose Blood Storage

- Francis Rous & James Turner add glucose (dextrose) solution
- Preserves fragile red blood cells from breaking down
- Citrated blood kept in ice chests remained viable up to 4 weeks
- Made storing blood possible in advance of military offensives

FOCUS CLUE
Why was glucose addition essential to prevent red blood cell breakdown?

3 **Nov 1917**

Cambrai Blood Bank Depot

- Capt Oswald Hope Robertson created world's first blood bank
- Collected universal Group O blood from healthy walking wounded
- Stored in 22 iced metal vacuum flasks at Casualty Clearing Station
- Treated 20 severely shocked soldiers at Cambrai; 11 survived

FOCUS CLUE
Why did pre-stored blood transform surgical survival during surprise assaults?

4 **1917–1918**

Harvey Cushing's Brain Surgery

- American surgeon Harvey Cushing pioneered delicate military neurosurgery
- Used local anaesthetic instead of general to prevent fatal brain swelling
- Silver clips clamped vessels (haemostasis); magnets extracted deep shrapnel
- Operated on 45 patients at 3rd Ypres, cutting mortality from 54% to 29%

FOCUS CLUE
Why was local anaesthesia superior [illegible] anaesthetic in head trauma?

5 **1917–1918**

Harold Gillies at Sidcup

- Harold Gillies founded Queen's Hospital, Sidcup, Kent (1917)
- Treated over 11,000 men with shattered faces and jaw trauma
- Pioneered tubed pedicle skin grafting to maintain active blood supply
- Grafts from chest/neck rebuilt noses and jaws; foundation of plastic surgery

FOCUS CLUE
How did the tubed pedicle technique prevent skin grafts from rotting and dying?

TIMELINE MISSION **Turn to Pages 2–3 (Milestone 6). Sketch Oswald Robertson's iced blood chest and Harold Gillies' tubed pedicle.** ← Pages 2–3

Robertson's Blood Depot: Ice chests, sodium citrate, and refrigerated blood at Cambrai—proof that cold beer isn't the only thing worth chilling.

KT5.6: Lifesaving Innovations: Blood Storage, Brain Surgery & Plastic Reconstruction

SOURCE A • PRIMARY MEDICAL ARTIFACT • CAMBRAI BLOOD DEPOT **ORIGIN: 1917**



The authentic wooden ice chest, glass collection flasks, rubber delivery tubing, and sodium citrate solutions utilized by Captain Oswald Robertson to establish the world's first blood bank during the Battle of Cambrai in November 1917.

Historical Provenance: Original portable blood transfusion outfit preserved in the Royal Army Medical Corps Historical Collection, London. Utilized during the Battle of Cambrai, November 1917.

• EDEXCEL SECTION A: QUESTION 1 FEATURE DRILLS [4 MARKS • 5 MINS] **2 SEPARATE QUESTIONS • 2 MARKS EACH**

Examiner Mark Scheme: Award 1 mark for identifying a valid historical feature + 1 mark for supporting factual detail. To score 2/2, the supporting detail must directly explain and develop the named feature.

1(a) Describe one feature of Harvey Cushing's surgical techniques in operating on head wounds on the Western Front. [2 marks]
Target: AO1 [2 marks] • Point (1m) + Detail (1m)
Sentence Stems: *One key feature was that... • Specifically, this was evident when...*

1(b) Describe one feature of the surgical methods used by Harold Gillies at Queen's Hospital, Sidcup. [2 marks]
Target: AO1 [2 marks] • Point (1m) + Detail (1m)
Sentence Stems: *One key feature was that... • Specifically, this was evident when...*

• QUESTION 2(B): 4-PROMPT FOLLOW-UP ENQUIRY [4 MARKS • 5 MINS] **AO3 HISTORICAL ENQUIRY**

Study Source A. How could you follow up Source A to find out more about the success of stored blood transfusions at the Battle of Cambrai in 1917? [4 marks]

Examiner Strategy: Follow up an exact quotation or visual detail. Your question must directly target your chosen detail. Your named source type must be authentic and specific (e.g. RAMC casualty returns, battalion war diaries).

Official Edexcel Prompt	AO3 Disciplinary Enquiry Strategy • Exam Standard
1. Detail in Source A to follow up:	*Select a precise, verifiable fact or visual detail directly related to the enquiry.*
2. Question I would ask:	*Frame a focused historical question that directly investigates your chosen detail.*
3. Type of source I would use:	*Name a specific contemporary record (e.g. RAMC war diaries, casualty logs, medical records).*
4. How this would help me:	*Explain what unique statistical or medical data this source provides to answer your question.*

Key Vocabulary: Karl Landsteiner • ABO blood groups • sodium citrate (Hustin 1914) • glucose preservative (Rous & Turner 1916) • Oswald Robertson • Battle of Cambrai (1917) • blood depot • Harvey Cushing • silver clips & surgical magnets • Harold Gillies • Queen's Hospital Sidcup • tubed pedicle graft

A key detail to follow up is... • The question I would ask is... • The specific historical source type needed is... • This source would help answer my question because...

SOURCE EVIDENCE, CONTEXTUAL KNOWLEDGE & DISCIPLINARY ANALYSIS NOTES:

FORMAL EXAMINATION PRACTICE **Write your formal answers on Pages 26–27 →**

• QUESTION 1: FEATURE DRILLS [2 X 2 = 4 MARKS] **Q1 Total: [/ 4]**

1(a) Describe one feature of Harvey Cushing's surgical techniques in operating on head wounds on the Western Front. **[/ 2]**

FEATURE: _____

DETAIL: _____

_____

1(b) Describe one feature of the surgical methods used by Harold Gillies at Queen's Hospital, Sidcup. **[/ 2]**

FEATURE: _____

DETAIL: _____

_____

• QUESTION 2(B): 4-PROMPT FOLLOW-UP MATRIX [4 MARKS • 5 MINS] **Score: [/ 4]**

Detail in Source A that I would follow up:	
Question I would ask:	
Type of source I would use:	
How this would help me find out more:	

[Disciplinary Commentary: Justify why this source type is authoritative for your historical enquiry]

Oswald Robertson stored blood in iced chests with sodium citrate and glucose at Cambrai in 1917, saving dying men from shock.

[Question 2 Sustained Analysis Continued • Contextual Knowledge • Conclusion]

Q2 Score: [/ 4] **Lesson Total:** [/ 8] **Teacher Feedback / Target:**

PUPIL NAME: _____ TARGET: _____ CURRENT GRADE: _____

ENQUIRY	ASSESSMENT COMPONENT FOCUS	PAGE	SCORE	TEACHER COMMENT & NEXT STEPS
KT5.1	**Q1:** Describe Two Features of Trenches / Arras [4m]	p. 6	/ 4	MARKING FEEDBACK • TARGETS:
	Q2(a): Source Utility Assessment (Trench Design) [8m]	p. 6–7	/ 8	
KT5.2	**Q1:** Describe Two Features of Trench Foot / Fever [4m]	p. 10	/ 4	MARKING FEEDBACK • TARGETS:
	Q2(b): 4-Prompt Follow-Up Matrix (Living Conditions) [4m]	p. 10–11	/ 4	
KT5.3	**Q1:** Describe Two Features of Shrapnel / Poison Gas [4m]	p. 14	/ 4	MARKING FEEDBACK • TARGETS:
	Q2(a): Source Utility Assessment (Artillery Trauma) [8m]	p. 14–15	/ 8	
KT5.4	**Q1:** Describe Two Features of Dressing Station / FANY [4m]	p. 18	/ 4	MARKING FEEDBACK • TARGETS:
	Q2(b): 4-Prompt Follow-Up Matrix (Stretcher Transport) [4m]	p. 18–19	/ 4	
KT5.5	**Q1:** Describe Two Features of Thomas Splint / X-Rays [4m]	p. 22	/ 4	MARKING FEEDBACK • TARGETS:
	Q2(a): Source Utility Assessment (Mobile X-Rays) [8m]	p. 22–23	/ 8	
KT5.6	**Q1:** Describe Two Features of Cushing / Gillies Surgery [4m]	p. 26	/ 4	MARKING FEEDBACK • TARGETS:
	Q2(b): 4-Prompt Follow-Up Matrix (Cambrai Blood Bank) [4m]	p. 26–27	/ 4	

• DIGITAL LEARNING & RETRIEVAL HUB • INTERACTIVE LESSONS & QUIZZES

Interactive Textbook • 10/20 Questions Per Enquiry

KT5.1	KT5.2	KT5.3	KT5.4	KT5.5	KT5.6
Trenches & Mud	Trench Illness	Trauma & Gas	Evacuation Chain	Splints & X-Rays	Blood & Plastic
LESSON HUB	LESSON HUB	LESSON HUB	LESSON HUB	LESSON HUB	LESSON HUB
MASTERY QUIZ	MASTERY QUIZ	MASTERY QUIZ	MASTERY QUIZ	MASTERY QUIZ	MASTERY QUIZ
Score: [/ 10]	Score: [/ 10]	Score: [/ 10]	Score: [/ 10]	Score: [/ 10]	Score: [/ 10]

Western Front Spec Complete: 16 marks in the bag, zero gangrene, and not a single complaint to the War Office.